

AI Engineer Career Roadmap

Personalized 3-Month Strategic Plan · Prepared via Chain-of-Thought Analysis

TARGET: AI ENGINEER · 90 DAYS

CURRENT POSITION

Working Professional · Full-stack web background

Existing skills: JavaScript, HTML, CSS, Node.js, React, SQL

TARGET GOAL

Transition into an **AI Engineer** role within **3 months**, leveraging existing JS/full-stack strength to specialize in applied AI & LLM engineering.

Skill Gap Analysis

Skill Area	Current Level	Required Level	Gap	Notes
Python	None	Strong	HIGH	Core language for AI/ML — must learn immediately
ML/AI Fundamentals	None	Working knowledge	HIGH	Focus on applied concepts, not deep theory
LLM APIs & Prompting	None	Strong	MEDIUM	Fast to learn given dev background
Vector DBs / RAG	None	Working knowledge	MEDIUM	Key for real-world AI apps
Backend/API (Node.js)	Strong	Working knowledge	LOW	Directly transferable
SQL / Data Handling	Strong	Working knowledge	LOW	Already sufficient
React/Frontend	Strong	Nice-to-have	LOW	Useful for AI product demos/UI

✳ Recommended Learning Plan

Weeks	Focus	Key Topics
Wk 1-2	Python for AI	Syntax, data structures, NumPy, Pandas, virtual envs
Wk 3-4	AI/ML Fundamentals	Supervised learning basics, scikit-learn, embeddings, vector similarity
Wk 5-7	LLM Engineering	OpenAI/Anthropic APIs, prompt engineering, function calling, LangChain
Wk 8-10	RAG & Vector DBs	Pinecone/Chroma, chunking, retrieval pipelines, embeddings search
Wk 11-12	Deployment & Portfolio	FastAPI backend, Docker, deploy AI app, polish GitHub/portfolio

Suggested Projects

Project	Stack	Online	Offline
AI Chatbot with RAG over docs	Python, LangChain, React UI	• Post weekly AI build logs on LinkedIn • Join AI Engineer Discord/Slack communities • Engage in r/MachineLearning,	• Attend local AI/ML meetups monthly • Reach out to 5 AI engineers/week for coffee chats • Contribute to 1 open-source AI repo
Resume/Job matcher using embeddings	Python, OpenAI API, Node.js API		
Full-stack AI SaaS demo	FastAPI + React + Vector DB		

Networking Strategy

Monthly Milestones

Month 1 ● Python fluency achieved ● ML fundamentals covered ● 1 small Python project shipped	Month 2 ● Built LLM-powered app w/ RAG ● Published 4+ LinkedIn posts ● 5+ networking conversations	Month 3 ● Portfolio + resume AI-focused ● Deployed capstone AI project ● Actively applying / interviewing
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⚡ Immediate Next Actions

1. Enroll in a Python fundamentals course and complete Day 1 today
2. Set up GitHub repo: "ai-engineer-journey" to track projects publicly
3. Follow 10 AI engineers on LinkedIn and start engaging with their content
4. Block 1 hour daily on your calendar for the next 90 days for AI learning

Generated using Chain-of-Thought Reasoning